

6 (a) (i) 10 A1

(ii)
$$\Delta E = E_4 - E_1 = \frac{hc}{\lambda}$$
$$= \frac{(6.63 \times 10^{-34} \times 3.00 \times 10^8)}{97.5 \times 10^{-9}}$$
$$= 2.04 \times 10^{-18} \text{ J}$$
 B1

$$= \frac{2.04 \times 10^{-18}}{1.60 \times 10^{-19}}$$
$$= 12.75 \text{ eV}$$

$$E_4 = E_1 + \Delta E = -0.85 \text{ eV}$$
 A1

(b) (i) $13.6 \text{ eV} = \text{work function} + eV_s$ M1
 $\text{work function} = 13.6 - 8.13$
 $= 5.47 \text{ eV}$ A1

(ii) max kinetic energy of photoelectrons = 8.13 eV B1

$$8.13 \times 1.60 \times 10^{-19} = \frac{p^2}{2 \times 9.11 \times 10^{-31}}$$
$$p = 1.5395 \times 10^{-24} = 1.54 \times 10^{-24} \text{ N s}$$
 A1

(iii)
$$p = \frac{h}{\lambda}$$
$$1.5395 \times 10^{-24} = \frac{6.63 \times 10^{-34}}{\lambda}$$
 B1

$$\lambda = 4.31 \times 10^{-10} \text{ m}$$
 A1

(iv) $\Delta p \Delta x \geq h$
 $9.11 \times 10^{-31} \times (1.2 \times 10^6 \times \frac{0.0025}{100}) \times \Delta x \geq 6.63 \times 10^{-34}$ B1

$$\Delta x \geq 2.43 \times 10^{-5} \text{ m}$$
$$\Delta x_{\min} = 2.2 \times 10^{-5} \text{ m (1 or 2 sf)}$$
 A1

5

- (v) Electrons are accelerated to high kinetic energy in a strong electric field. B1
- when a stream of high energy electrons are rapidly brought to rest (decelerated) by collisions with the atoms of tungsten, radiation in the form of X-rays are emitted.
 - If the electron loses only part of its initial kinetic energy during the collision with the target atom, the photon emitted has energy equal to the loss in kinetic energy of the bombarding electrons. B1
 - Since the loss in kinetic energy can take any value, through the multiple stages of deceleration of electrons in a metal target, a continuous X-ray spectrum is obtained. B1